

have a service; to send emails, we have a service; there is a service for push notifications, and then we have services to communicate with all these services. So, in all, we have a lot of services. Also, we have integration with a lot of cinemas and payment gateways. So we are completely a services model. Some of those services are based on legacy systems, as the project started as early as 2005-06, though we launched it in 2007. However, quite a lot of them were created over a period of time and newer services are coming up, day in and day out. All the new services that we are developing are on open source platforms. But, some of the legacy applications are still on .NET and we maintain them like that.

There is no hard and fast policy for us to move from .NET to open source technologies, but wherever it makes sense, we do so. One of the key reasons for us to migrate is scalability. It is much easier to scale on open source technologies. There are so many tools freely available for that in the open source domain. In .NET, you have to stick to the processes that are created by one company. Because of such limitations, in many cases, we choose to go with open source technologies.

Q Apart from scalability, what other reasons motivate you to migrate to open source technologies?

Flexibility, which is very important to us, is one other reason. There are options available in open source that are backed up by a company. When you are in business, you need some sort of assurance. So if you look at MongoDB, the company is also there to back you up in case you need some support. But at the same time, open source gives you a lot of flexibility—the whole process of how you want to do things is completely in your control. Automation is another key aspect. The way we deploy our servers, everything is automated; whereas in Windows, we still have to keep on clicking the 'Next' option, after every step.

Implementation of the automation program on Windows is much more time consuming for us, and with such a small percentage of our operations running on Microsoft systems, it does not make sense any more. We have about 40 servers running Windows, vis-à-vis over 600 servers that run on Linux. So, even if there are automation tools available for Windows, it does not make sense for us to implement them and they will not be compatible with the Linux tools. We are investing a lot in Linux tools for automation, deployment and the whole continuous delivery platform. So, where Windows works for us, we make use of it, but not always.

Q Did you face any issues when you introduced open source technology in your company?

What we have done so far is that we have limited the use of open source technology only to the engineering team. From the business side, most of the people use Windows on their desktops for their day-to-day work. This is followed even in the case of applications; for instance, in the IPL, the box

office team runs the point-of-sale system that sells tickets, which in turn runs on Windows. The application they use still runs on .NET. But when the application does a transaction on the server, it happens on Linux.

Essentially, we have put all the open source technologies in the data centre layer, which is exclusive to the engineering team. So from the company's point of view, we do not face any hurdles or a brick wall, when things like training is required. When it comes to the engineering team, when we hire engineers, we make sure that they are trained in open source technologies.

Q Do you get enough talent that is trained in open source technologies?

We get a lot of talent that is already trained in open source technologies. However, a lot of talent is home grown. We hire a lot of freshers, who have just come out of engineering colleges and even institutes like NITT, and we train them. We look at a candidate with an open mind, because for us, it is not a choice between open source and Windows; it is more about what works best in business use cases. It's just that in most of the cases, open source gives us more flexibility, because of which we choose it.

We are not biased towards any platform. As a business, we cannot be biased. Any business looks at flexibility and a good turnaround time while choosing any technology. And the best way to attain that goal is to use a hybrid model. So, if I become biased and say that all my systems, even for the end users, will use open source, training those hundreds of people will become a nightmare, and it involves a lot of money too. In a company like ours, we have a lot of temporary staff, particularly during events like the IPL. Thousands of people join us on a temporary basis. Training them for a short period can be a nightmare.

The fact is that Windows is something that everybody is aware of. Except for engineering colleges, all other colleges and schools still use Windows. It had become the default for them. So, one can assume that they are aware of it. This is why we use a hybrid model and it definitely helps us. In future, if people start using open source operating systems like Ubuntu, we will have no issues in adopting it on the user level as it will save a lot of training costs for us. If that happens, we will be happier because we will not end up paying so much on licensing fees.

Q You said that you have a lot of temporary employees; but you must be having a lot of permanent employees as well. Did you ever try using open source technology on a user-level? What was the outcome?

We had tried this experiment a few years back. I wouldn't say that it was a failure, but it definitely was not a success. We told our employees to try something different, but we did not get a lot of positive acceptance. But when we look back at what happened, we realise we now have a large number of